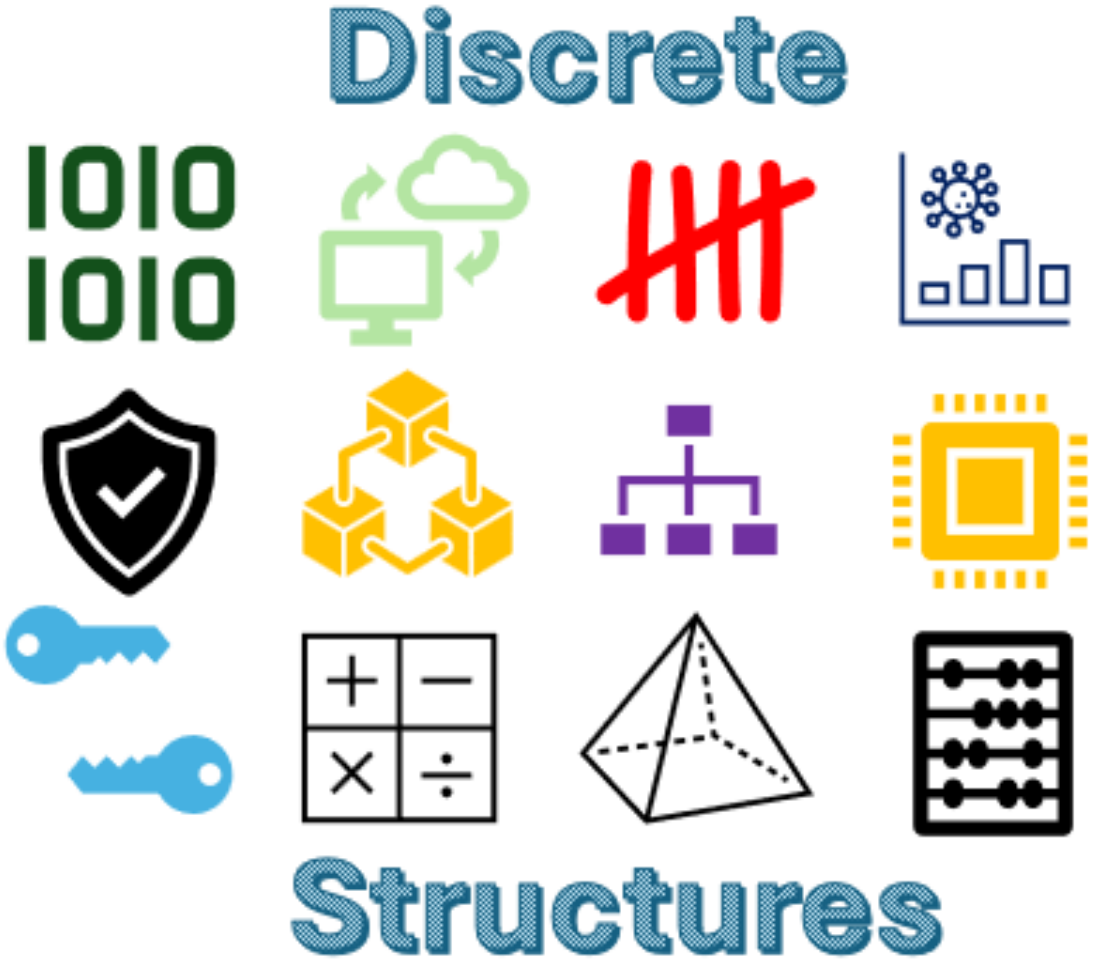
COMP 2310: Discrete Structures
Fall 2026

Chapter 1: Fundamentals



Matrices

- Boolean matrix operation

Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

(a) Compute $\mathbf{A} \vee \mathbf{B}$. **(b)** Compute $\mathbf{A} \wedge \mathbf{B}$.

Solution

(a) Let $\mathbf{A} \vee \mathbf{B} = [\, c_{ij} \,]$. Then, since a_{43} and b_{43} are both 0, we see that $c_{43} = 0$. In all other cases, either a_{ij} or b_{ij} is 1, so c_{ij} is also 1. Thus

$$\mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

(b) Let $\mathbf{A} \wedge \mathbf{B} = [\, d_{ij} \,]$. Then, since a_{11} and b_{11} are both 1, $d_{11} = 1$, and since a_{23} and b_{23} are both 1, $d_{23} = 1$. In all other cases, either a_{ij} or b_{ij} is 0, so $d_{ij} = 0$. Thus

$$A \wedge \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Matrices

- Boolean product

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$. Compute $\mathbf{A} \odot \mathbf{B}$.

Solution

Let $\mathbf{A} \odot \mathbf{B} = [\, e_{ij} \,]$. Then $e_{11} = 1$, since row 1 of $\mathbf{A}$ and column 1 of $\mathbf{B}$ each have a 1 as the first entry. Similarly, $e_{12} = 1$, since $a_{12} = 1$ and $b_{22} = 1$; that is, the first row of $\mathbf{A}$ and the second column of $\mathbf{B}$ have a 1 in the second position. In a similar way we see that $e_{13} = 1$. On the other hand, $e_{14} = 0$, since row 1 of $\mathbf{A}$ and column 4 of $\mathbf{B}$ do not have common 1's in any position. Proceeding in this way, we obtain

$$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

Mathematical structures

- **Mathematical structures:** Collection of objects with defined operations and accompanying properties.
- **Three major components:**
  - **Object:** elements of the structures: set or matrices
  - **Operations:** action performed on the objects: union, intersection
  - **Properties:** attributes of the operations: closures, commutativity, associativity, distributivity.

Attributes of operations: Closures

- A structure is closed with respect to an operation if performing that operation on any two objects in the structure results in another object within the same structure.
- Example of a closed structure

Example 3 The structure $(5 \times 5 \text{ matrices}, +, *, ^T)$ is closed with respect to addition because the sum of two 5×5 matrices is another 5×5 matrix. ◆

- Binary operation: requires two objects
- Unary operation: requires one objects
- Example of a non-closed structure

Example 4 The structure (odd integers, $+$, $*$) is not closed with respect to addition. The sum of two odd integers is an even integer. This structure does have the closure property for multiplication, since the product of two odd numbers is an odd number. ◆

Attributes of operations: Commutative

- **Commutative:** Order does not matter

(a) Join and meet for Boolean matrices are commutative operations.

$$\mathbf{A} \vee \mathbf{B} = \mathbf{B} \vee \mathbf{A} \quad \text{and} \quad \mathbf{A} \wedge \mathbf{B} = \mathbf{B} \wedge \mathbf{A}.$$

- **Non-commutative:** Order matters

(b) Ordinary matrix multiplication is not a commutative operation. $\mathbf{AB} \neq \mathbf{BA}$. ◆

Attributes of operations: Associativity

- **Associative:** Grouping among three or more objects does not matter
  - Requires to have the same operation

$$(x \square y) \square z = x \square (y \square z).$$

Attributes of operations: Distributive

- **Distributive:** distribution of one operation over another

$$x \square (y \nabla z) = (x \square y) \nabla (x \square z).$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Mathematical structures: De-Morgan's Law

- **First Law**: The complement of the union of two sets is equal to the intersection of their complements.

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

- **Second Law:** The complement of the intersection of two sets is equal to the union of their complements.

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

- Note that:
  - $\overline{A}$ = U – A
  - $\overline{AUB}$ = U – $\overline{AUB}$